IRP Tsuyu

Week 1

Project Plan 12th June.

The main focus will be increasing the spatial resolution of wildfire satellite images. Step 1 of the big project. If we have time we can go to step 2.

Paper to follow:

https://www.mdpi.com/2072-4292/16/4/715

https://www.sciencedirect.com/science/article/abs/pii/S092427162100160X?via%3Dihub
(30m spatial resolution)

1. GOES vs **MTG**. If it's in Google earth engine it's easier to collect data. What bands are we using? Etc
2. Sentinel-2(20m) vs Landsat(30m) what do we want?
3. Cascade approach or direct approach, or both and compare

Coarse Input Data: GOES vs MTG

	GOES	**MTG**
Spatial Res	2km	Visible and NIR: 1km IR: 2km (Active Fire Monitoring EO:EUM:DAT:0682)
Temporal Res	10 mins	Every 10 mins
Bands	3.9μm	FCI channel IR-3.8μm
Spatial coverage	The Americas (North and South)	Lat: -67.5° to 67.5°, Lon: -67.5° to 67.5° (EU, Africa, Atlantic Ocean, parts of Middle East and South America?)

Temporal extent	GOES-16: May 2017 - Now GOES-17: 2018 - now GOES-18: Oct 2022 - Now	Jul 2025 - Now
Access	GEE API	EUMETSAT API
Advantages	Can access via GEE API	Directly relevant to the ESA project. Greater spatial extent
Limitations	Only covers the Americas	Data only available from July 2025
Relevant links	GEE GOES-16 FDCF Data Guide	Data guide

- Since GOES likely will have more wildfire data (or could check if the spatial breadth of the MTG compensates for the short time period), we could initially work with GOES as our coarse input data. If time permits, we can try tuning the model with MTG?
- Alternatively we can combine MTG and MSG data to get a larger dataset for MTG (provided that the relevant data have the same spatial resolution. The IR bands of MSG are 3km while 1/2km for MTG.
- For MTG, we find that some bands have different resolution (0.5, 1, 2km) due to the different modes (High Resolution Fast Imagery (HRFI))
- Fire Radiative Power - MTG have data available in 1km from May 2026

High-res Target Data: Landsat vs Sentinel-2

	Landsat	**Sentinel-2**
Spatial res	30m	20m
Temporal res	8/16 days	5 days
Relevant Bands	Band 6 (1.6 μm), Band 7 (2.2 μm)	Band 11 (1.6 μm), Band 12 (2.2 μm)
Temporal extent	Landat 8: Apr 2013 - present Landsat 9: Sept 2019 - present In NASA FIRMS: June	March 2017 - present

	2022 - present	
Spatial extent	NASA FIRMS: US/Canada only	global
Access	GEE	Copernicus Data Space Ecosystem (API)
Advantages	Contains thermal infrared band Fire dataset is available	Slightly higher resolution
Limitations	NASA FIRMS only have data from 2022, and only available for US/Canada	Lacks thermal infrared bands. As far as I can find, there is no clear 'fire' dataset that I could use

- Active fire detection in Landsat-8 imagery: A large-scale dataset and a deep-learning study Only fire data for Aug/Sept 2020 (worldwide).

Intermediate Resolution Data

- VIIRS:
  - Resolution: 375m
  - Access via NASA FIRMS
  - Temporal Res: Daily
- Main problem: when doing the cascaded approach, we need to spatially and temporally match VIIRS with the high-res target (landsat or Sentinel). There could potentially be a temporal mismatch

Summary: Likely use GOES for input coarse data, VIIRS for intermediate, and Landsat (Pereira et al. dataset) for 'ground truth'

Main Research

I have chosen to compare the following architectures:

Experiment	Architecture	Pipeline Structure	Input Resolution	Output resolution	Model type
E1	EDiffSR	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	Diffusion

E2	EDiffSR	Cascaded	Stage 1: 2km -> 375m Stage 2: 375m -> 30m	30m (Landsat, Pereira et al.)	Diffusion
E3	HAUNet	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	Attention UNet
E4	HAUNet	Cascaded	Stage 1: 2km -> 375m Stage 2: 375m -> 30m	30m (Landsat, Pereira et al.)	Attention UNet
E5	ESTNet	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	Transformer
E6	ESTNet	Cascaded	Stage 1: 2km -> 375m Stage 2: 375m -> 30m	30m (Landsat, Pereira et al.)	Transformer

So for the Cascade approach, first you train a model that goes from 2km to 375m. Then we use that model to increase the res of Goes so we can match Goes and Landsat.

WEEK 2

Objectives:

1. Only satellites or more data? (Landcover, DEM, …). Please select the bands we are using in the satellites.
2. Evaluation metric (what we use to know if the result is good, MSE? Something that measures the blurriness)
3. Decide 3 architectures to try.
4. Ask gemini for a plan to train the model in Google Cloud Platform. So ask gemini what is the best way to ingest the data (data pipeline big query ? or?). IRP plan

Inputs/outputs/possible covariates:

- **GOES-16/17/18** bands (from https://www.sciencedirect.com/science/article/pii/S2666017221000031):
    - Active fire bands: 3.80–3.99 µm and 10.82–11.60 µm at 2km
    - Alternatively, use Level 2 Fire Mask (FSH) or Fire Radiative Power (FRP) bands directly, at 2 km.
- **VIIRS** (375m):
    - Use NASA FIRMS data, which uses I-bands: Bright_ti4, Bright_ti5, frp (fire radiative power) for thermal anomaly detection.

- **Topography:** Copernicus DEM GLO-30 (30m)
- **Landcover:** ESA WorldCover 10m v100 **(10m) (Here it tells you the water bodies)**
- **NDWI (Normalised Difference Water Index):** Landsat Collection 2 Tier 1 Level 2 32-Day NDWI Composite **(30m) (dont need this)**
- **NDVI (Normalised Difference Vegetation Index):** Landsat Collection 2 Tier 1 Level 2 8-Day NDVI Composite **(30m)**
- **Global inland water: GLCF:** Landsat Global Inland Water **(30m)**
- **Buildings: GHSL:** Global settlement characteristics (10 m) 2018 (P2023A) **(10m)**
- **Landsat** (Pereira et al. dataset) (30m)
    - Uses 10 bands: Channels b1 to b11 except b8: Green, Red, Near-Infrared (NIR), Shortwave Infrared 1 (SWIR 1), Shortwave Infrared 2 (SWIR 2), Cirrus, Thermal Infrared 1 (TIRS 1), Thermal Infrared 2 (TIRS 2)

Metrics

- IOU/F1 (binary mask):
    - IOU Quantifies the overlap between the predicted fire mask and the ground truth mask.
    - F1: The harmonic mean of precision (how many predicted fires were real) and recall (how many real fires were detected).
- IPSNR

17 June 2026

1. **Choose an appropriate machine to download the data and preprocess it and filter it. Save the name of the machine + costs, and add that to your final report.**
2. **See if we have enough images with at least (more than 10 pixels)**
3. **GOES from VIIRS (rasterised).**

Total target Landsat (Pereira et al.) masks (North America 1, North America 2, South America, and manual masks): 10,012
After running pipeline and filtering data where number of active pixels is >= 10: 1214 Landsat dataset pairs. Can consider reducing the threshold.

Pereira et al. Landsat fire dataset contains images of 256x256 at 30m (physical extent 7680m). This means the corresponding GOES image of the same area at the same physical extent will result in a 4x4 pixel image. Two options:

1. Get GOES data that covers the same physical extent (4x4 pixel) for consistency with the target image.
2. Get GOES data that also covers the surrounding area (so the image covers greater physical extent) for more information for the model.

Loss: Combination of dice and focal loss

1 July 2026

1. **Try to have 10k images. Start with VIIRS downscaling (from GOES to VIIRS). Train model and see results.**
2. **Explore https://www.usgs.gov/landsat-missions/landsat-collection-2-level-3-burned-area-science-product?utm_source=chatgpt.com**

More sources to collect more landsat data

3. **Try to have 10k**

Project Progress

Data acquisition & pipeline

- GOES 16 and 17 (2km): Used channel 7 and 14 (fire sensitive and background channels) and TFI (calculated from channel 7 and 14). Acquired via public bucket in Google Cloud: GOES 16, GOES 17. Initially, GOEs was acquired via GEE but due to it being very slow, we migrated to acquiring from the public bucket instead.
- Landsat (30m): Acquired from Google drive by Prereira et al. Paper: "Active fire detection in Landsat-8 imagery: A large-scale dataset and a deep-learning study"
- Covariates (both acquired using GEE API):
    - Copernicus DEM
    - ESA Landcover
- VIIRS (375m): downloaded from NASA FIRMS Archive

Overall pipeline:

- We need datasets with the following:
    - VIIRS/Covariates/GOES pairs
    - Landsat/Covariates/GOES pairs
- For each dataset, I acquired the GOES and covariates so they cover the same physical extent and same geographical area, and ensure GOES are matched temporally with the target.
- The pipeline was built and run inside Google Cloud Platform (GCP). Any finances needed for using the virtual machine on GCP is covered by Artio). GCP was used due to better computers, and also the advantage of not running the pipeline locally (which involves tons of data and running locally on my computer will run into many problems). Using GCP also allows me to store tons of data into the cloud inside the GCP bucket. Using cloud computing aligns with industry standards.
- Preprocessing includes excluding any data where a high fraction of the GOES patch contains 0 (e.g. due to sensor malfunction etc.), excluding target Landsat where active pixels are < 10.
- Data splitting is done by binning the target datasets based on the number of active pixels, and applying a stratified shuffle split so data in train/test/validation are balanced.

- The dataset pipeline is pretty much completed. If there's enough time, can consider integrating weather data, and frames (e.g. using previous fire frames to help model to see places that are already burnt) but unlikely considering speed of progress.
- All essential code is packaged as a python script rather than written in jupyter notebooks so Artio can use them in the future.

Main challenges:

- Unfamiliarity with using a lot of the tools, such as Google Cloud Platform, GEE API. Also unfamiliar with many of the data engineering problems and concepts. Since Jorge was also not an expert in using GCP, I had to figure out most of it myself, such as how to set up a GCP instance to run my code, and how to save the data acquired into my GCP bucket etc.
- Spent quite a significant time debugging the code, such as:
- Pipeline kept crashing due to unbounded cache
    - Fix: Set a max storage limit and when the cache hits near the limit, it deletes the recently accessed files to free up space
- Tried implementing threading to speed up process, but led to segfaults
    - Fix: simplified the architecture to a single-threaded loop to guarantee reliability.

Downscaling Models

Direct: GOES (2km) -> Landsat (30m)
Cascaded: GOES -> VIIRS (375m) -> Landsat

The model used are are all Deep Learning models

Models in progress:

- Hybrid Attention UNet (HAUNet): Similar to UNet architecture except:
    - Uses spatial and channel attention blocks in their encoder/decoder.
    - Uses cross-scale interaction module (CIM), which is used as skip connection between different scale feature outputs of encoders to bridge the semantic and resolution gaps between them. (see figure below)
- Simple Autoencoder baseline based on Badhan et al. paper

Losses:

- Attempted to implement the following: Focal Loss, Dice loss, focal dice loss, and Gaussian focal loss (based on the loss used in CenterNet)

Proposed evaluation metrics:

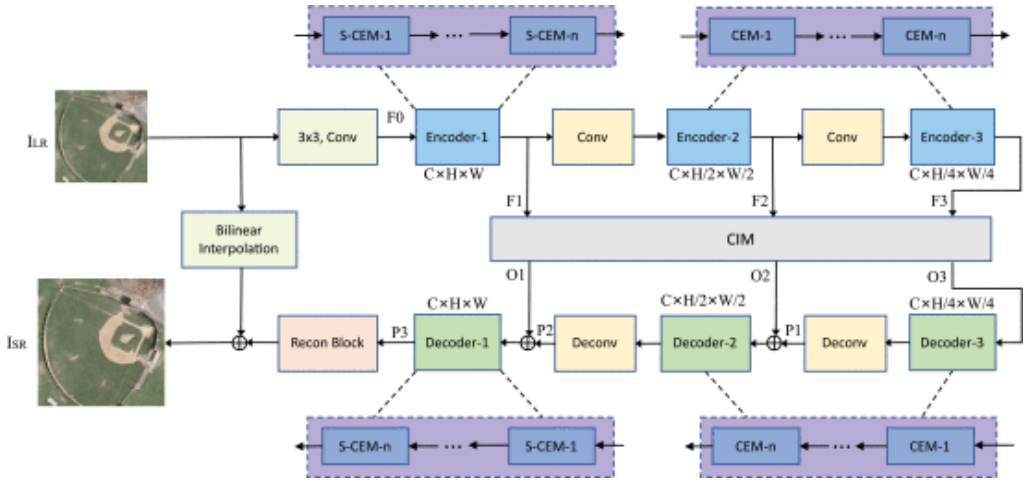
- Intersection of Union, boundary F1 score, precision and recall.

Main challenges:

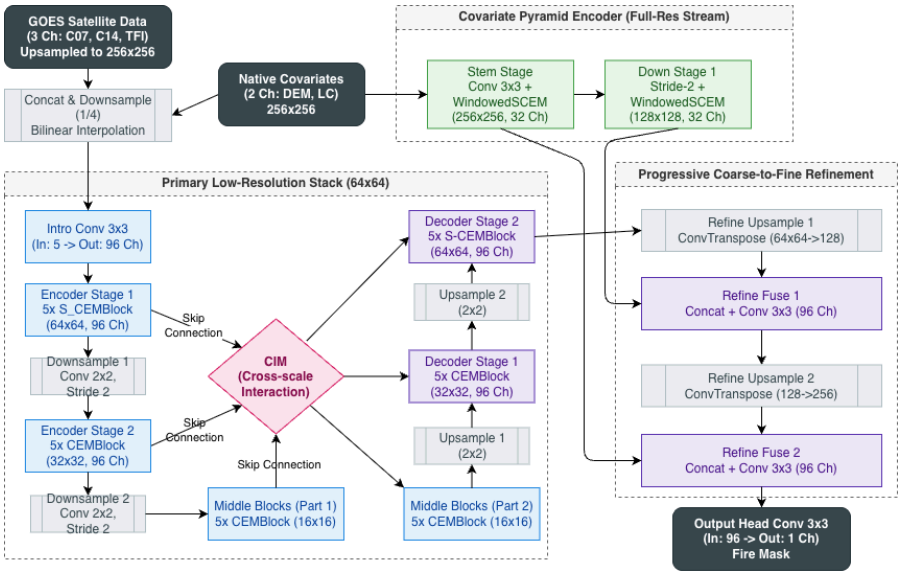
- The original HAUNet uses full spatial attention to learn global context, which is fine for small images (e.g.64x64) but for larger ones (we have covariate data of sizes 256x256)

this ran into OOM issues. To keep the full attention, the input data is downsampled to 64x64. However this means that the high-resolution covariates are processed at lower resolution, meaning we lose some high-res info. To ensure the model still processes the full high-res covariates, I built a covariate encoder that runs in parallel to the main HAUNet. The covariate encoder uses windowed spatial attention to avoid the massive computational cost needed to process them at full resolution. At the refinement decoder, the outputs from the covariate encoder are progressively fused in.

The original HAUNet architecture:



The modified HAUNet for this project:

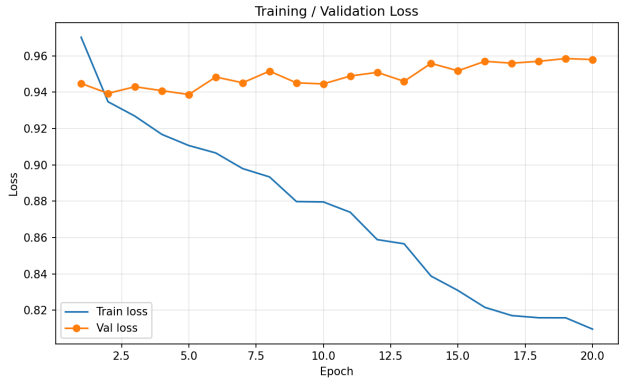


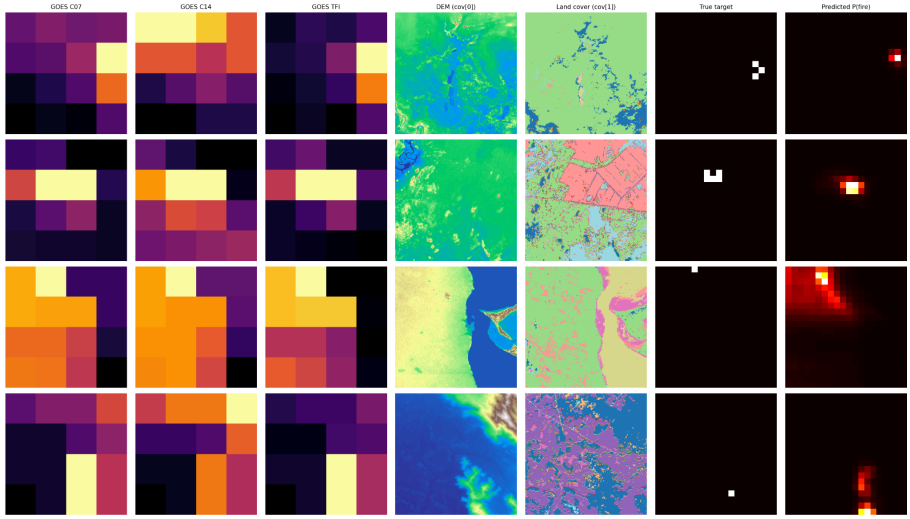
- The target datasets (VIIRS and Landsat) are very imbalanced, where most is background (many contain only a couple of active pixels). Choosing an ordinary loss like MSE/RMSE will cause the model to produce an output where it predicts everything as ‘no fire’. Hence I have implemented various losses to attempt to handle this: Focal, Dice, Focal&Dice, Gaussian Focal losses (I am yet to compare the results of these different losses).

Next steps:

- Finish implementation of the HAUNet and the autoencoder baseline, and train the model on full dataset for both cascaded and direct downscaling. If time allows, start implementing a third model.

Cascaded model GOES->VIIRS (Focal Dice Loss):
Train: 7100 samples / 3550 batches
Val : 888 samples / 444 batches
Best AUPRC: 0.131868

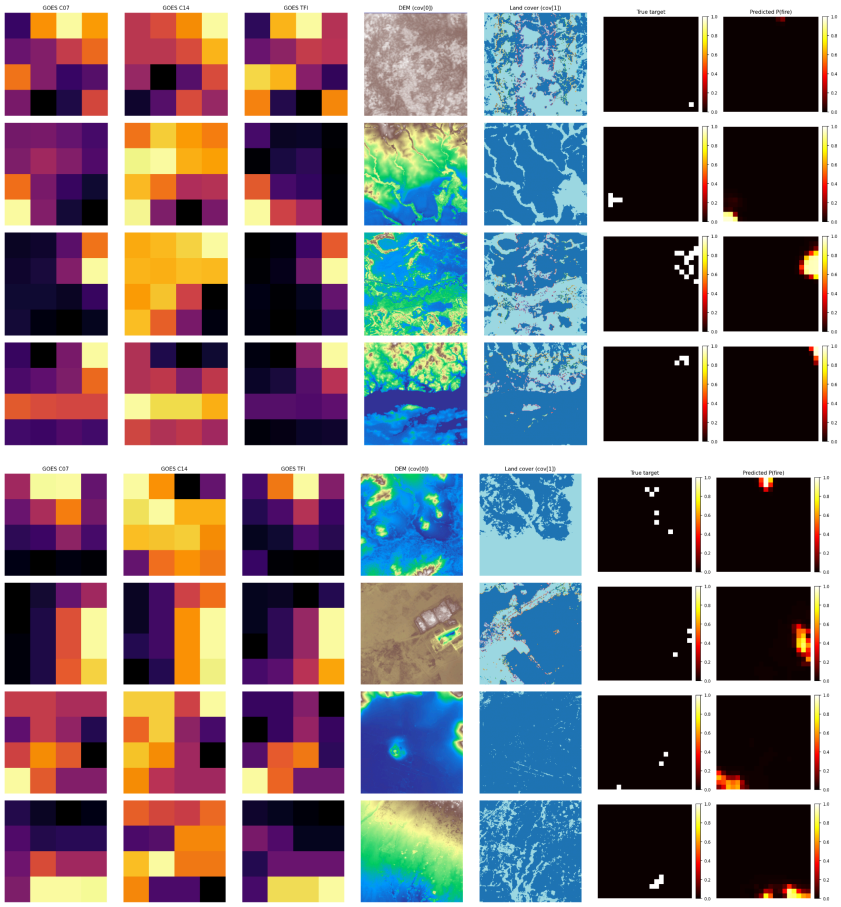


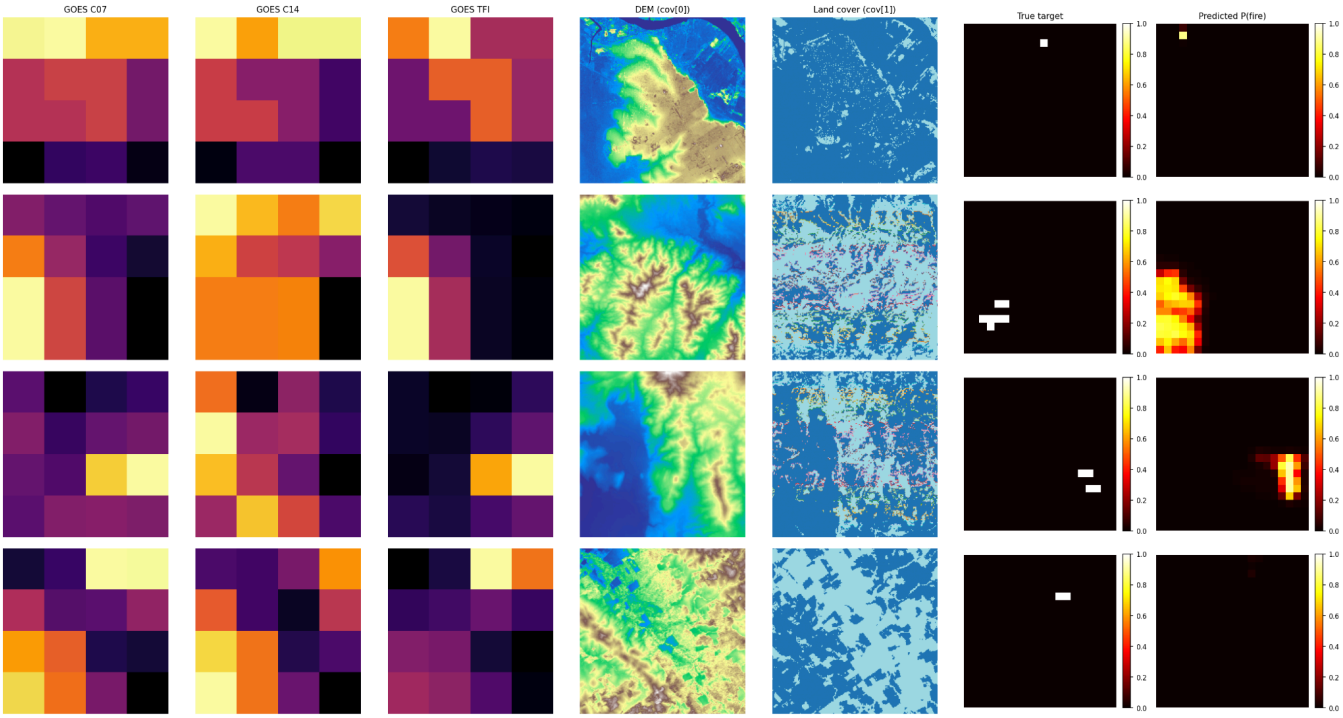


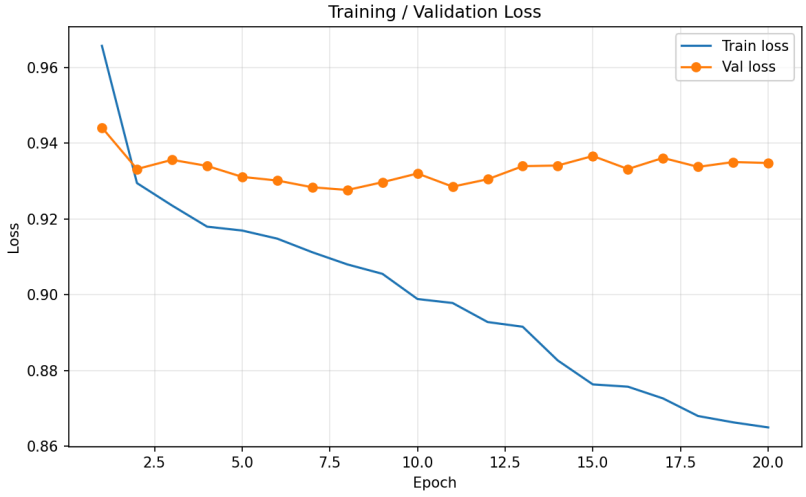
After adding dropout, augmentation, and landcover onehotencoding. (Best AUPRC: 0.139476) (v2):

Metric	Mean	Std	Median
iou	0.0626	0.1094	0.0000
soft_iou	0.0597	0.0985	0.0031
boundary_f1	0.0962	0.1597	0.0000
auprc	0.1399	0.1905	0.0667

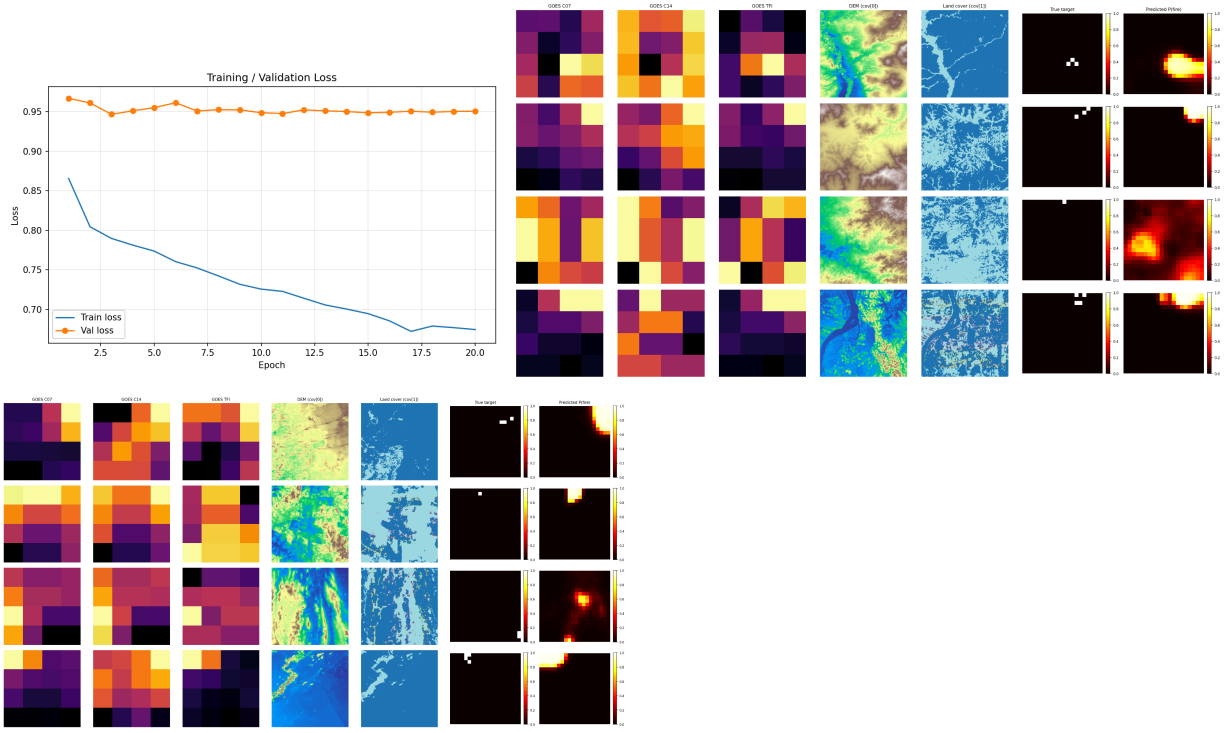
precision	0.0733	0.1272	0.0000
recall	0.2215	0.3430	0.0000
pixel_f1	0.1012	0.1630	0.0000



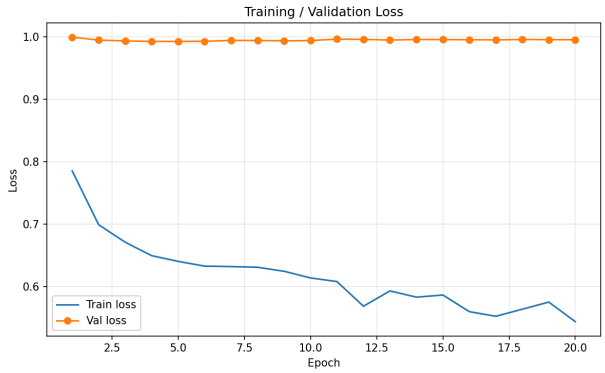




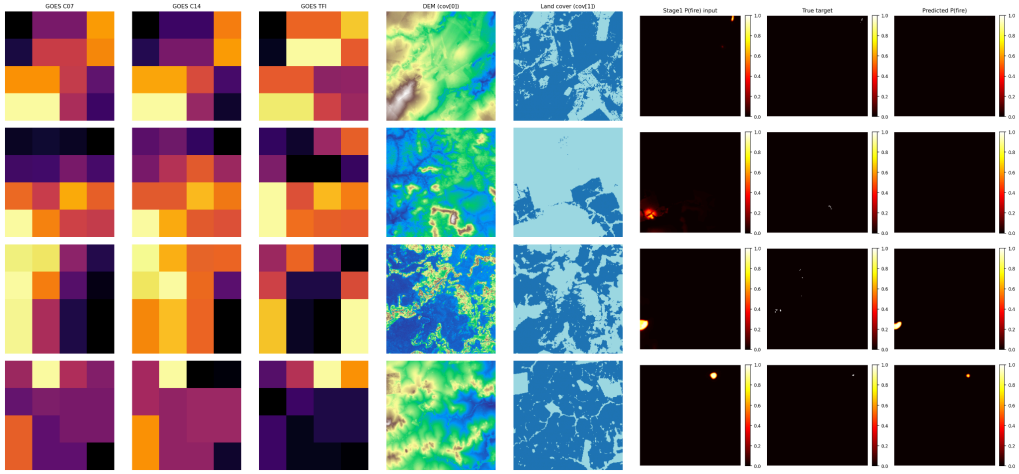
After dilating targets (keeping onehotencoding, augmentation, dropout) (v3):
Best val_auprc: 0.1381



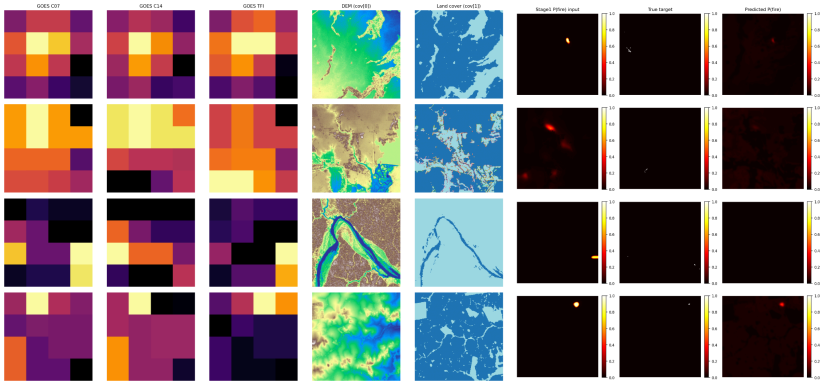
Cascaded stage 2 (VIIRS->Landsat):



With best val_soft_iou (0.0063):

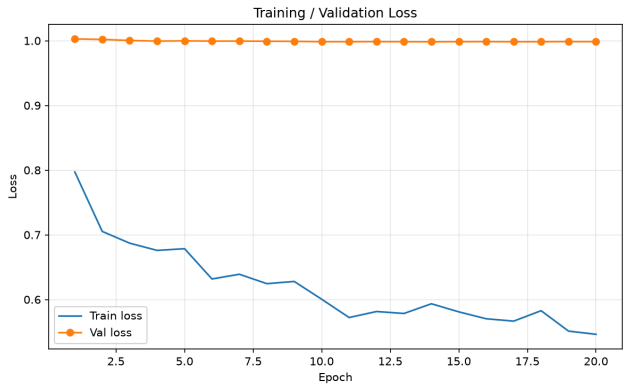


With best val_auprc (0.0193):

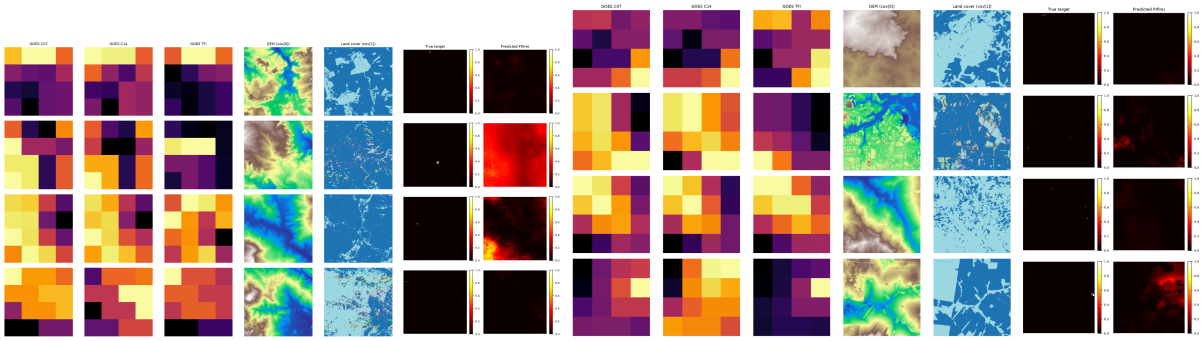


Best model checkpoints are selected based on the prioritised metrics
Metrics to prioritise:

HAUNet Direct focal dice (with cov pyramid encoder, no target dilation):

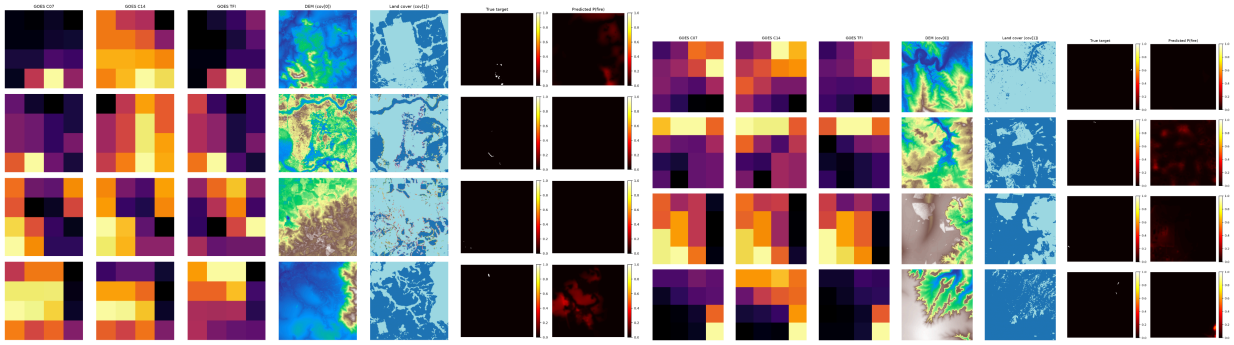


Best val_auprc=0.0174:



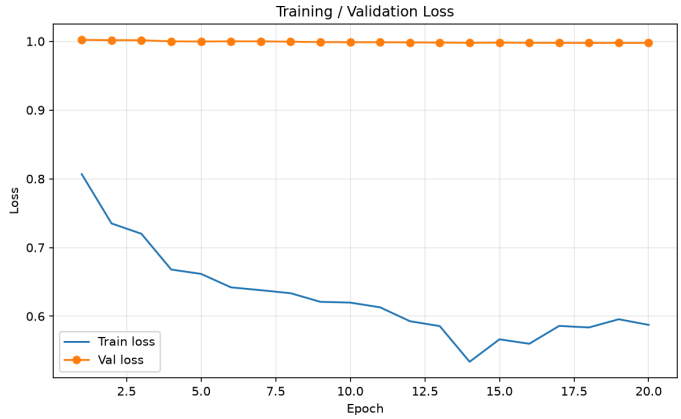
Metric	Model	AllZero	Prevalence
soft_iou	0.0013	0.0000	0.0002
boundary_f1	0.0004	0.0000	0.0000
auprc	0.0103	0.0006	0.0006
precision	0.0004	0.0000	0.0000
recall	0.0069	0.0000	0.0000
pixel_f1	0.0007	0.0000	0.0000

Best val_soft_iou=0.0024:

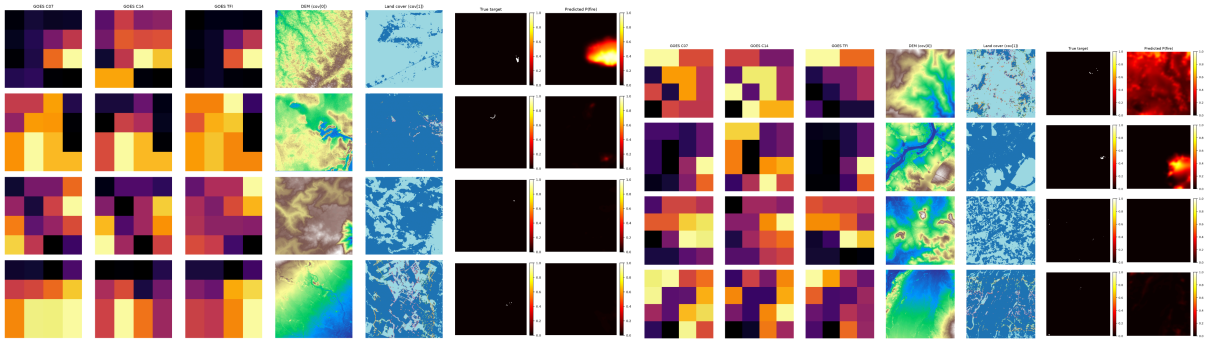


Metric	Model	AllZero	Prevalence
soft_iou	0.0013	0.0000	0.0002
boundary_f1	0.0004	0.0000	0.0000
auprc	0.0133	0.0006	0.0006
precision	0.0006	0.0000	0.0000
recall	0.0121	0.0000	0.0000
pixel_f1	0.0011	0.0000	0.0000

HAUNet Direct focal dice, no cov encoder

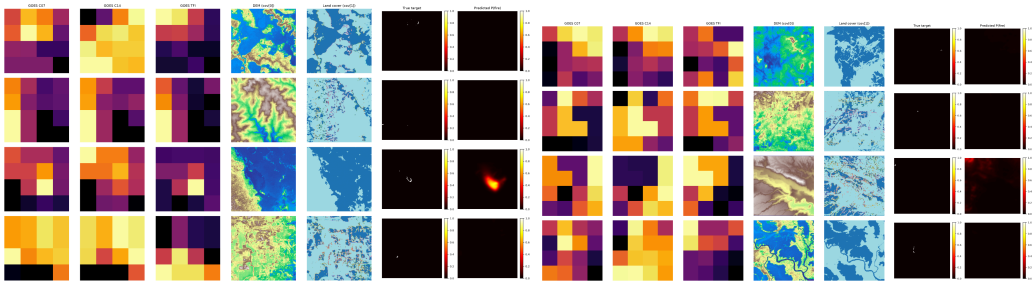


Best val_soft_iou=0.0025:



Metric	Model	AllZero	Prevalence
soft_iou	0.0024	0.0000	0.0002
boundary_f1	0.0014	0.0000	0.0000
auprc	0.0096	0.0006	0.0006
precision	0.0032	0.0000	0.0000
recall	0.0210	0.0000	0.0000
pixel_f1	0.0048	0.0000	0.0000

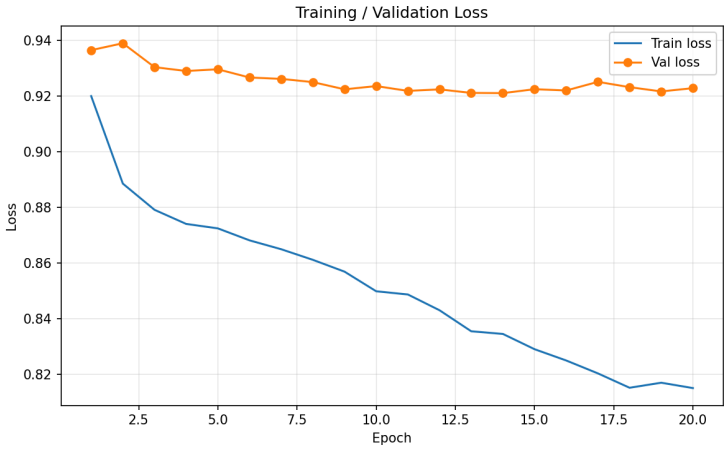
Best val_auprc=0.0160



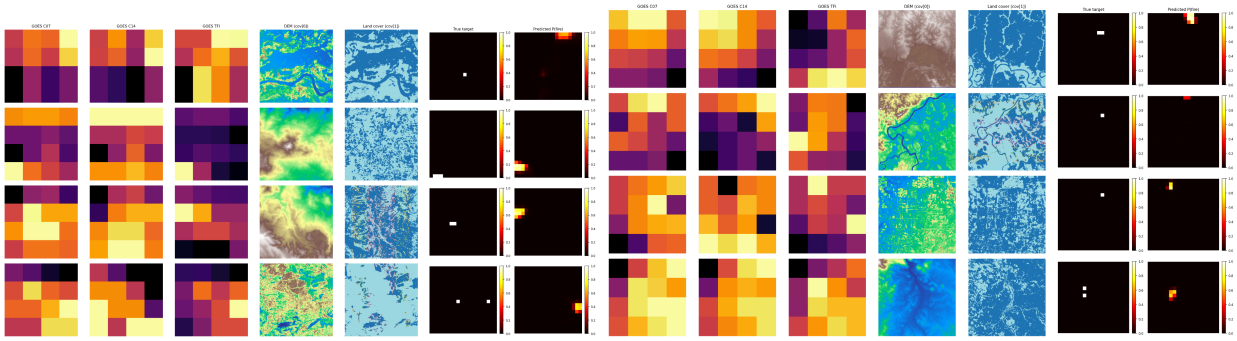
Metric	Model	AllZero	Prevalence
soft_iou	0.0024	0.0000	0.0002

boundary_f1	0.0014	0.0000	0.0000
auprc	0.0096	0.0006	0.0006
precision	0.0032	0.0000	0.0000
recall	0.0210	0.0000	0.0000
pixel_f1	0.0048	0.0000	0.0000

HAUNet cascaded stage 1 (GOES->VIIRS) focal dice:

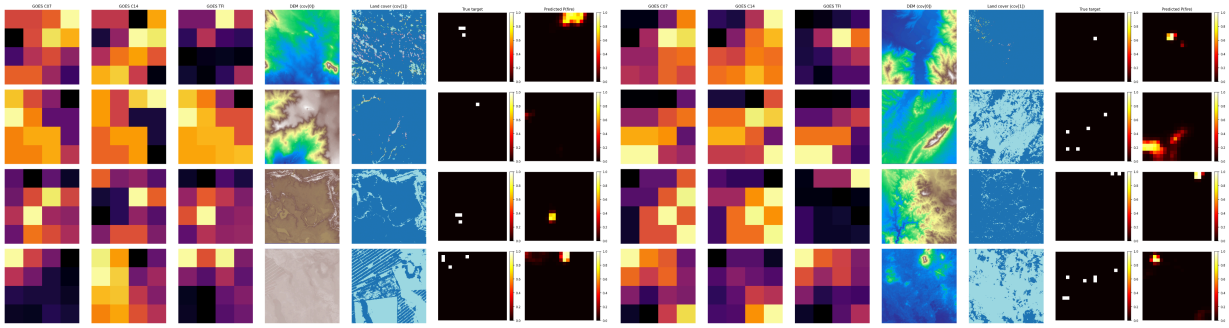


Best val_soft_iou=0.0625:



Metric	Model	AllZero	Prevalence
soft_iou	0.0625	0.0000	0.0035
boundary_f1	0.1010	0.0000	0.0000
auprc	0.1435	0.0082	0.0082
precision	0.0765	0.0000	0.0000
recall	0.2617	0.0000	0.0000
pixel_f1	0.1075	0.0000	0.0000

Best val_auprc=0.1487:



Metric	Model	AllZero	Prevalence
soft_iou	0.0604	0.0000	0.0035
boundary_f1	0.0983	0.0000	0.0000
auprc	0.1458	0.0082	0.0082
precision	0.0739	0.0000	0.0000
recall	0.3006	0.0000	0.0000
pixel_f1	0.1087	0.0000	0.0000

HAUNet cascaded stage 1 (GOES->VIIRS) focal dice, no cov encoder: